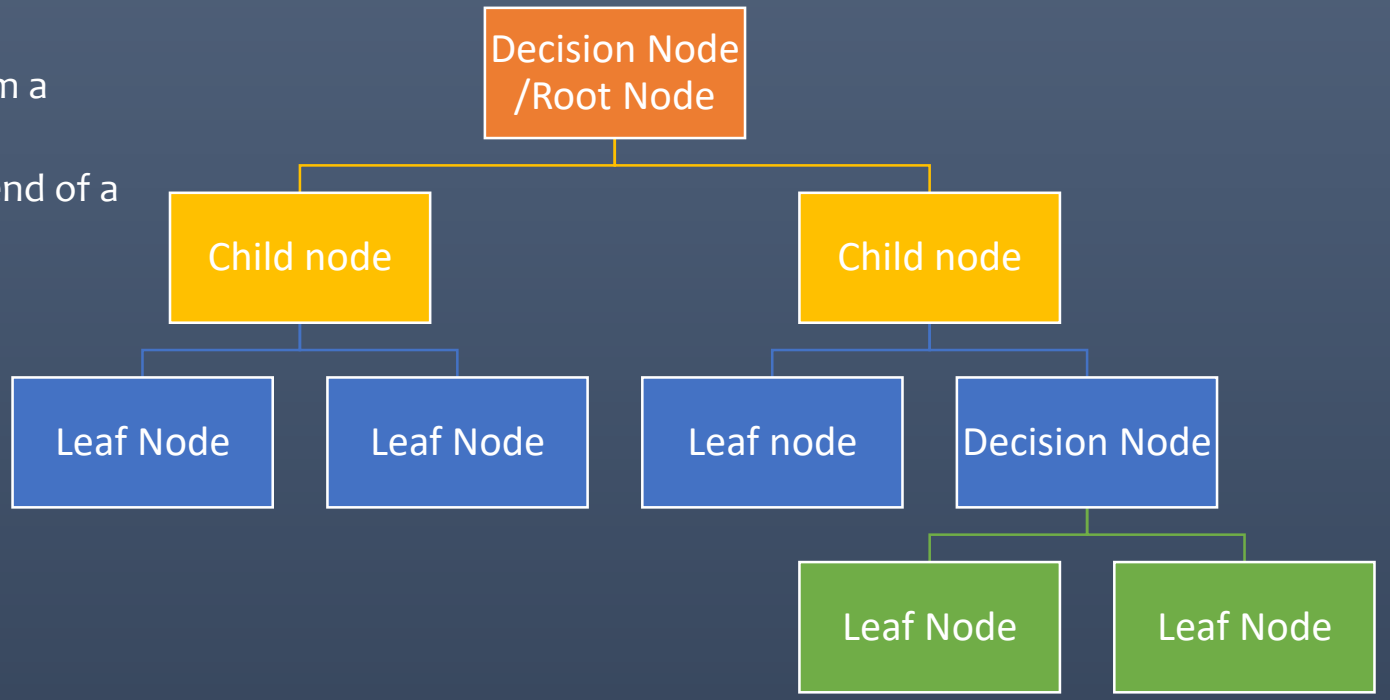
Decision Tree

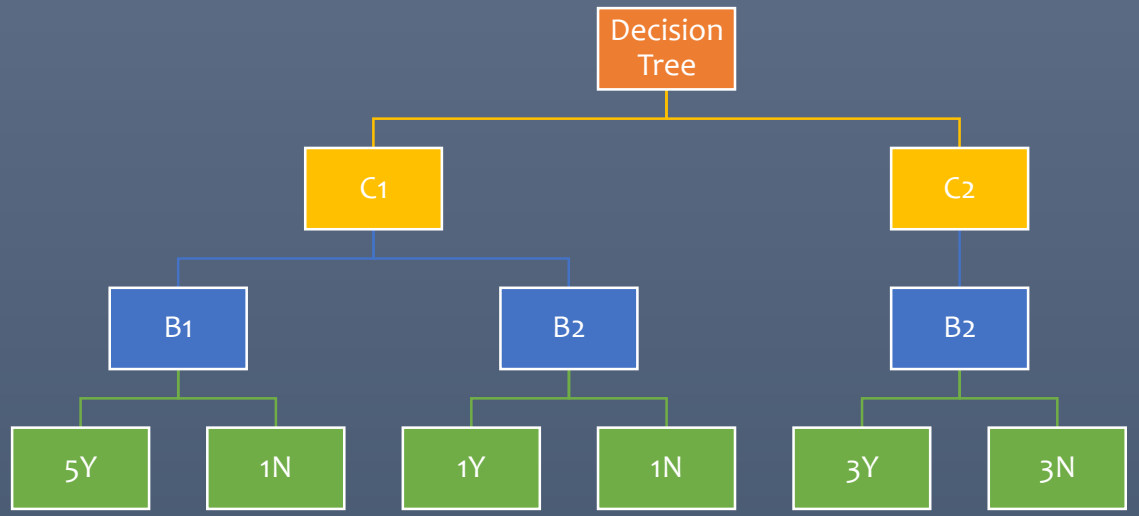
This is a tree-based approach and A Decision Tree is a Supervised learning technique that can be used for classification and Regression problems,

however, it is mostly preferred for solving Classification problems. It is a tree-structured classifier, where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome

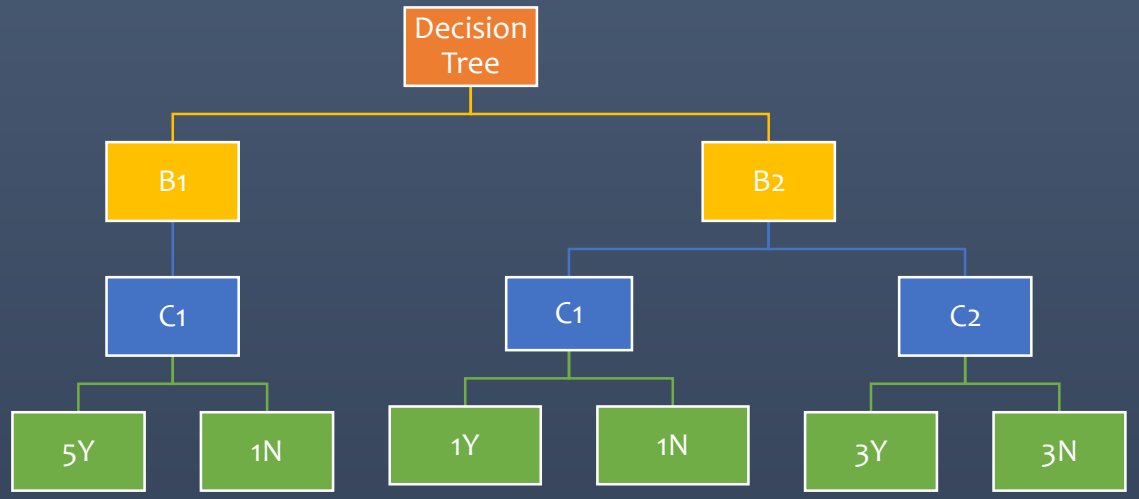
Root nodes: The top-level nodes in a tree structure from which all other nodes descend.
Child nodes: Nodes that are directly connected and descend from a parent node.
Leaf nodes: Nodes that have no children and are located at the end of a tree branch.



Feature 01	Feature 02	Yes/No
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B1	Y
C1	B2	Y
C2	B2	Y
C2	B2	Y
C2	B2	Y
C1	B1	N
C1	B2	N
C2	B2	N
C2	B2	N
C2	B2	N



When constructing the tree, the model must choose one feature among multiple features as the parent node. This selection is based on the concept of **information gain**, where the feature with the highest information gain becomes the parent node, while the remaining features become the child nodes.



It is basically **splitting the data into a tree-like structure**.
A Decision Tree has three main parts:

- **Root Node** – the starting point of the tree.
- **Child Nodes** – the branches created after splitting the data.
- **Leaf Nodes** – the final result or prediction.

Since there are multiple features in the dataset, we need to decide **which feature should be the Root Node.**

There are tow DT

- **DTC(Information gain)**
- **DTR(SDR)**